Sensor HAT v3 Universal — HAT for Raspberry Pi or Seeed XIAO host | SCD41 CO2 | SHT40 T/RH | SGP41 VOC/NOx | VEML7700 light | ICS-43434 mic

PWR_FLAG PWR_FLAG PWR_FLAG

Power: Pi 5 V or XIAO USB 5 V via ideal diodes -> +5V -> AP2112K-3.3 LDOs, enabled by Pi 3V3 GPIO4

I2C bus: VEML7700 0x10, SHT40 0x44, SGP41 0x59, SCD41 0x62 (4.7k pull-ups; the Pi adds 1.8k)

XIAO socket J7/J8 with 100R series resistors. NEVER fit a XIAO while the board sits on a Pi

Raspberry Pi header (female socket, bottom side)

I2S microphone: GPIO18 BCLK, GPIO19 LRCLK, GPIO20 DIN / XIAO D1, D0, D2

ID EEPROM (Pi 3V3; WP high = protected, bridge JP2 to program)

1-Wire (GPIO4 / XIAO D5 via JP6)

UART (GPIO14/15 / XIAO D6/D7) + GPIO17: series R, ESD, J3 pin 1 = 3.3 V (JP1) or switched/fused 5 V (JP3), never both

Test points and mounting holes

Particulate-matter port: own LDO, JP4 (closed) = 3.3 V for SEN6x, or cut JP4 + bridge JP5 = 5 V for SEN5x

5 V ideal-diode OR (LM66100 x2)

VOC/NOx SGP41 (0x59)

Expansion: Qwiic (3.3 V I2C) + GPIO pads (Pi only)

Button GPIO27 / XIAO D4

Status LED (GPIO10 / XIAO D10, 5 V via 74LV1T125) and buzzer (GPIO13 / XIAO D3, PWM)

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Sheet	/
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